Voiding Dysfunction

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Block: Endocrine/Urology/Reproduction

Thread: Clinical Medicine

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Further Reading

Clare J. Fowler, Derek Griffiths, William C. De Groat. The neural control of micturition. *Nature Reviews Neuroscience*. 2008;9(6):453-466. doi:10.1038/nrn2401 (View Online)

Panicker, Jalesh N, Fowler, Clare J, Kessler, Thomas M. Lower urinary tract dysfunction in the neurological patient: clinical assessment and management. *The Lancet Neurology*. 2015;14(7):720-732. doi:10.1016/S1474-4422(15)00070-8 (View Online)

Learning Objectives

By exam time, you should be able to do the following:

1. Identify the components of the lower urinary tract and describe their functional role.
2. Describe the pathways for lower urinary tract sympathetic, parasympathetic and somatic innervation.
3. Explain the difference between stress, urgency, and overflow incontinence.
4. Describe the nature of lower urinary tract dysfunction based on the location of a neurologic lesion.
5. Describe etiologies for bladder outlet obstruction and identify how they are treated.

Study Questions

1. The majority of sympathetic innervation of the lower urinary tract comes from:
    a. pelvic plexus
    b. hypogastric nerve
    c. pudendal nerve
    d. Onuf's nerve
    e. Alcock's plexus

2. Is the following statement regarding the micturition/voiding cycle true or false:

The first step is detrusor contraction and urethral smooth muscle relaxation. The second step is voluntary relaxation of the striated external urethral sphincter.

Lower Urinary Tract Anatomy

A. Components
    1. Bladder
    2. Urethra

B. Bladder Body
    1. Allows dynamic filling/emptying
    2. Smooth muscle bundles are oriented in random directions

C. Bladder Outlet
    1. Trigone
        a. Location of ureteral insertion and drainage
        b. Smooth muscle bundles in longitudinal and circular orientation
    2. Urethra
        a. Internal urethral sphincter = bladder neck
            i. Not a true circumferential "sphincter" muscle
            ii. Composed of smooth muscle
            iii. Under involuntary control (provides passive continence)
        b. External urethral sphincter

i. Composed of smooth and striated muscle

ii. Under involuntary and voluntary control (provides passive and active continence)

D. Natal Female Urethra

1. 3-5 cm in length

2. Short urethral length predisposes these individuals to urinary incontinence and UTIs compared to those identified as male at birth

E. Natal Male Urethra

1. Length is variable (approximately 20 cm)

2. Long urethral length protects against urinary incontinence and UTIs compared to those identified as female at birth

3. Proximal urethra passes through prostate gland – prostatic enlargement can impede the flow of urine

F. Bladder Blood Supply

1. Anterior trunk of internal iliac artery gives rise to:

a. Superior vesical artery – supplies bladder body and distal ureters

b. Inferior vesical artery – supplies bladder base, distal ureters, prostate and seminal vesicles

i. Inferior vesical artery = vaginal artery in women

G. Natal Female Urethral Blood Supply

1. Anterior trunk of internal iliac artery gives rise to:

a. Internal pudendal artery

b. Inferior vesical artery (vaginal artery)

H. Natal Male Urethral Blood Supply

1. Anterior trunk of internal iliac artery internal pudendal artery dorsal artery of the penis

2. Blood supply also arises from corpus spongiosum

I. Histology

1. Urothelium is comprised of 3-7 layers of transitional cells

a. Function: impermeable barrier to urine, allows for expansion/contraction

J. Lower Urinary Tract Function

1. Storage of urine

a. Low pressure storage protects kidneys (high bladder pressures are transmitted to kidneys and can lead to renal function impairment)

b. Maintain continence

c. Provide appropriate sensation and warning time to get to a bathroom

2. Emptying

a. Should be under voluntary control

b. Requires coordination between bladder detrusor muscle and urethral sphincters

c. Empty to near completion

Normal Lower Urinary Tract Physiology

A. Micturition (voiding) Reflex – requires integration of autonomic and somatic nervous system

1. First step is voluntary relaxation of the striated external urethral sphincter

2. Second step is involuntary simultaneous detrusor contraction and urethral smooth muscle relaxation

B. Spinal Cord Centers

1. T10-L2 – sympathetic input via the hypogastric nerve

2. S2-S4 – sacral micturition center

a. Parasympathetic nucleus derives cell bodies from the pelvic nerve

b. Onuf's nucleus (pudendal nucleus) derives cell bodies from the pudendal nerve

C. Peripheral Nerves – all have both motor and sensory function

1. Hypogastric nerve

a. Sympathetic fibers → sympathetic chain ganglia → pelvic plexus

2. Pelvic nerves

a. Parasympathetic S2-S4 ventral roots → pelvic plexus

3. Pudendal nerve

a. S2-S4 ventral roots

b. Somatic - controls striated urethral and anal sphincters

D. Lower Urinary Tract Receptors

1. Sympathetic innervation

a. Beta-3 adrenergic receptors – relax bladder

b. Alpha adrenergic receptors – contract bladder neck and urethra

2. Parasympathetic innervation

a. Acetylcholine (ACH) receptors – contracts detrusor, primarily mediated by M2>M3 receptors

b. Non-adrenergic/non-cholinergic receptors – mediated by ATP and nitric oxide

E. Somatic Innervation

1. Mediated by pudendal nerve

2. Onuf's nucleus (pudendal nucleus) – controls striated muscle of external urethral sphincter

3. Centrally located motor nucleus controls striated muscle of pelvic floor

F. Sensory Innervation

1. Ad fibers – located in detrusor muscle, sense bladder distension, bladder inflammation leads to increased afferent firing at lower threshold

2. Unmyelinated C fibers – located in urothelium, sense bladder distension at higher threshold (when bladder is over-distended), responds to noxious stimuli (e.g. chemical irritants, cold)

G. Lower Urinary Tract Pharmacologic Targets

1. Anti-cholinergics (aka anti-muscarinics) – block ACH mediated detrusor contraction and reduce afferent activity of Ad and C fibers

2. Beta adrenergic agonist – inhibits detrusor contraction, beta-3 is predominant receptor in human bladder

3. Onabotulinum toxin (Botox®) - blocks presynaptic release of ACH, inhibits detrusor contraction, reduces afferent activity

Urinary Incontinence

A. Definition: involuntary loss of urine

B. Epidemiology

1. Hart to determine accurate prevalence rates since many do not seek care

2. Prevalence increases with age

3. Prevalence rates significantly higher in women compared to men

Stress Urinary Incontinence

A. Definition: urinary incontinence with physical activity or Valsalva maneuver (e.g. cough, laugh, sneeze)

B. Pathophysiology

1. Urethral resistance overcome by sudden increase in abdominal pressure
2. Mechanisms
    a. Intrinsic sphincter deficiency – loss of urethral mucosal coaptation
    b. Urethral hypermobility – loss of peri-urethral support, makes it easier to overcome urethral resistance

C. Risk factors: increased age, obesity, smoking, chronic cough, connective tissue disorder, family history, previous pelvic reconstruction surgery, pregnancy (women), previous prostate surgery (men)

D. Treatment: meant to increase peri-urethral support (increase bladder outlet resistance)

1. Pelvic floor muscle exercises (e.g. Kegels)
2. Vaginal insert for female patients
3. Peri-urethral bulking agents
4. Urethral sling
5. Artificial sphincter for male patients

Overactive Bladder + Urgency Urinary Incontinence

A. Definitions

1. Urinary urgency – sudden compelling desire to urinate which is difficult to defer
2. Urinary frequency – complaint of voiding more often during waking hours than previously deemed "normal" (patient dependent)
3. Urgency urinary incontinence – incontinence associated with urgency ("can't make it to the bathroom in time")

B. Pathophysiology: involuntary detrusor contractions and/or increased afferent input

1. Etiology is most often idiopathic but certain neurologic disorders are associated with urinary urgency and urgency incontinence (e.g. Multiple sclerosis, CVA)

C. Treatment

1. First line – behavioral modifications
    a. Fluid management, reduction of bladder irritants (e.g. caffeine)
    b. No associated risks, should be offered to all patients
2. Second line – pharmacologic therapy (anti-cholinergics, beta-3 agonist)
3. Third line – bladder neuromodulation
    a. Peripheral tibial nerve stimulation
    b. Sacral neuromodulation
    c. Chemodenervation using Onabotulinum toxin
4. Fourth line – bladder augmentation with a patch of bowel

Overflow Incontinence

A. Definition: urinary incontinence as a result of bladder over-distension (bladder just cannot hold any more urine)

B. Clinical Presentation

1. Urinary retention is a prerequisite
2. Diagnosed by imaging (ultrasound, CT scan, MRI) or catheterization

C. Risk factors: bladder outlet obstruction, poor detrusor muscle contractility, medications (e.g. alpha agonists)

D. Treatment: goal is to improve bladder emptying

1. Indwelling catheter
    a. Pros: minimal maintenance
    b. Cons: UTI risk, requires exchange every 4 weeks
2. Intermittent catheterization (catheter only placed long enough to drain the bladder)
    a. Pros: reduces UTI risk, patients not physically attached to a catheter with drainage bag
    b. Cons: need to have motivated patient with good manual dexterity, must be performed 4+ times per day

Neurogenic Bladder

A. Definition: dysfunctional lower urinary tract secondary to a brain, spinal cord or peripheral nerve lesion

1. Nature of lower urinary tract dysfunction depends on location of neurologic lesion

2. Acute lower urinary tract dysfunction may differ from the chronic dysfunction

B. Lesions above the brain stem (e.g. cerebrovascular accident [CVA], brain tumor)

1. Loss of central inhibition of micturition/voiding cycle

2. Involuntary detrusor contractions lead to overactive bladder and urgency incontinence

3. CVA patients may initially present with urinary retention secondary to an acontractile detrusor

C. Spinal cord lesions (e.g. spinal cord injury, spinal cord tumor)

1. Spinal shock – initial period of hypocontractile or acontractile detrusor that leads to urinary retention

a. Lasts approximately 6 weeks following a spinal cord injury

2. Involuntary detrusor contractions lead to overactive bladder and urgency incontinence

D. Lower motor neuron lesion (e.g. nerve injury during radical pelvic surgery, cauda equine syndrome)

1. Acontractile detrusor leads to urinary retention

E. Demyelinating lesions (e.g. Multiple sclerosis)

1. Involuntary detrusor contractions lead to overactive bladder and urgency incontinence

F. Diabetes

1. Diabetic peripheral neuropathy can affect bladder innervation

2. Clinical presentation is variable

a. Involuntary detrusor contractions lead to overactive bladder and urgency incontinence

b. Hypocontractile or acontractile detrusor that leads to urinary retention

Bladder Outlet Obstruction

A. Definition: increased resistance at outlet leading to impaired bladder emptying

1. Impaired bladder emptying can range from complaint of weak urinary stream to complete urinary retention

B. Primary bladder neck obstruction: failure of involuntary internal urethral sphincter to relax during voiding

1. Treatment – a receptor antagonist

C. Dysfunctional voiding: failure of voluntary external urethral sphincter to relax during voiding

1. Treatment – pelvic floor muscle biofeedback teaches patients to control pelvic floor muscles and voluntarily coordinate sphincter relaxation with detrusor contraction

D. Bladder outlet obstruction in those identified as female at birth

1. Pelvic organ prolapse

a. Cystocele leads to urethral kinking and impaired bladder emptying

b. Treatment – correct cystocele

2. A urethral sling placed to treat stress incontinence can be too tight and lead to bladder outlet obstruction

a. Treatment – remove portion of the urethral sling

E. Bladder outlet obstruction in those identified as male at birth

1. Urethral stricture

a. Treatment – remove scar tissue or place a catheter to drain the bladder

2. Prostatic enlargement – benign prostatic enlargement is more common than prostate cancer

a. Medical therapy

i. Alpha receptor antagonist – relaxes smooth muscle at bladder neck

ii. Phosphodiesterase-5 inhibitor – leads to relaxation of smooth muscle at bladder neck

iii. 5-alpha reductase inhibitor – blocks conversion of testosterone to dihydrotestosterone and subsequently reduces prostate size

b. Surgical therapy – prostate reduction procedure or complete prostate removal for prostate cancer treatment

c. Catheter placement is also appropriate for benign prostatic enlargement

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. *Hypogastric nerve*

2. *False*